1. （1） 在极坐标系中，$L=\frac{1}{2}m(\dot{r}^2+r^2\dot{\theta}^2)-V(r)$

不显含θ，所以对应的广义动量为守恒量

$$P_\theta=\frac{\partial L}{\partial\dot{\theta}}=mr^2\dot{\theta}=\text{常数}$$

这个P_θ就是角动量。

$$[q_\alpha,\ q_\beta]=0$$

（2）基本泊松括号：$[p_\alpha,\ p_\beta]=0$

$$[q_\alpha,\ p_\beta]=\delta_{\alpha\beta}=\begin{cases}0 & (\alpha\neq\beta)\\ 1 & (\alpha=\beta)\end{cases}$$

$$\frac{df}{dt}=\frac{\partial f}{\partial t}+\sum_\alpha\left(\frac{\partial f}{\partial q_\alpha}\dot{q}_\alpha+\frac{\partial f}{\partial p_\alpha}\dot{p}_\alpha\right)$$

正则方程（或哈密顿方程）$\begin{cases}\dfrac{dq_\alpha}{dt}=\dfrac{\partial H}{\partial p_\alpha}\\ \dfrac{dp_\alpha}{dt}=-\dfrac{\partial H}{\partial q_\alpha}\end{cases}$ 代入上式得：

$$\begin{aligned}\frac{df}{dt}&=\frac{\partial f}{\partial t}+\sum_\alpha\left(\frac{\partial f}{\partial q_\alpha}\frac{\partial H}{\partial p_\alpha}-\frac{\partial f}{\partial p_\alpha}\frac{\partial H}{\partial q_\alpha}\right)\\ &=\frac{\partial f}{\partial t}+[f,\ H]\end{aligned}$$

（3）

$$\begin{aligned}&L=\sum_{i=1}^{3n}\frac{1}{2}m\dot{x}_i^{\,2}-V(x_i)\\ &\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{x}_i}\right)-\frac{\partial L}{\partial x_i}=0\\ &\frac{d}{dt}(m\dot{x}_i)+\frac{\partial V(x_i)}{\partial x_i}=0\\ &m\ddot{x}_i=-\frac{\partial V(x_i)}{\partial x_i}\\ &\quad=F_i\end{aligned}$$

即牛顿运动方程

（4）哈密顿作用量 $s=\int_{t_1}^{t_2}L(q,\dot{q},t)dt$， L=T-V 为有势系拉氏量

哈密顿原理：有势力系在通过位形空间中两个点所能作的各种（相邻）运动中，真实运动使哈密顿作用量取极值。

对于单摆：$L = T - V = \frac{1}{2}ml^2\dot{\theta}^2 - (-mgl\cos\theta)$

$$= \frac{1}{2}ml^2\dot{\theta}^2 + mgl(1-\frac{1}{2}\theta^2)$$

$$= \frac{1}{2}ml^2\dot{\theta}^2 - \frac{1}{2}mgl\theta^2 + mgl$$

$$= \frac{1}{2}ml^2\theta_0{}^2\frac{g}{l}\cos^2\sqrt{\frac{g}{l}}t - \frac{1}{2}mgl\theta_0{}^2\sin^2\sqrt{\frac{g}{l}}t + mgl$$

$$= \frac{1}{2}mgl\theta_0{}^2(\cos^2\sqrt{\frac{g}{l}}t - \sin^2\sqrt{\frac{g}{l}}t) + mgl$$

$$= \frac{1}{2}mgl\theta_0{}^2\cos 2\sqrt{\frac{g}{l}}t + mgl$$

故

$$s = \int_0^T Ldt$$

$$= \frac{1}{2}mgl\theta_0{}^2\int_0^T \cos 2\sqrt{\frac{g}{l}}dt + mglT$$

$$= 0 + mglT$$

$$= mglT$$

（5） 满足正则方程 $\begin{cases}\frac{dq_\alpha}{dt} = \frac{\partial H}{\partial p_\alpha} \\ \frac{dp_\alpha}{dt} = -\frac{\partial H}{\partial q_\alpha}\end{cases}$ ， q_α，p_α 称为正则变量

变换 $\begin{cases}Q_\alpha = Q_\alpha(q,p,t) \\ P_\alpha = P_\alpha(q,p,t)\end{cases}$ 使得 Q_α, P_α 与 q_α, p_α 一样满足正则方程，则为正则变换。

正则变换的目的是：通过正则变换获得更多的循环坐标，从而得到更多的守恒量。

哈密顿—雅可比方法的实质：寻找这样一个母函数，使其由此所得的正则变换能使所有的新坐标 Q 和新动量 P 均不出现在哈密顿函数 $\tilde{H}$ 中，（或 $\tilde{H} \equiv 0$），从而 Q 和 P 均为常数，由母函数确定的变换方程将直接给出正则方程的解

$$\begin{cases}\frac{dQ_\alpha}{dt} = [Q_\alpha, \tilde{H}] = 0 \\ \frac{dP_\alpha}{dt} = [P_\alpha, \tilde{H}] = 0\end{cases}$$

2. 解： 重心 $\overrightarrow{oc}=\vec{a}$

重力矩 $\vec{L}=\vec{a}\times m\vec{g}$

$\dfrac{d\vec{G}}{dt}=\vec{L}$ （$\vec{G}$ 为陀螺角动量）

$\because\ \vec{L}\perp\vec{G}$ （因为高速自转， 所以 $\vec{G}\approx I_z\omega\vec{k}$ ）

$\therefore$ $\vec{G}$ 大小不变，只改变方向。

设进动角速度为 $\dot{\psi}$ （方向为 $o\xi$ 轴）

$\vec{G}$ 大小不变， oz 方向，随陀螺一起的 $\dot{\psi}$ 进动

$\therefore\ \dfrac{d\vec{G}}{dt}=\vec{\psi}\times\vec{G}$

$\therefore\ \vec{a}\times m\vec{g}=\vec{\psi}\times\vec{G}$

$amg\sin\theta=\dot{\psi}G\sin\theta$

$\therefore\ \dot{\psi}=\dfrac{mga}{G}=\dfrac{mga}{I_z\omega}$

3. 解： $F=\dfrac{mc}{r^3}$

$V(r)=-\int F dr$

$=\dfrac{mc}{2r^2}$ （取 ∞ 处势能为零）

机械能守恒： $\dfrac{1}{2}mv_0{}^2=\dfrac{1}{2}mv^2+\dfrac{mc}{2a^2}$

有心运动角动量守恒

$\vec{b}\times m\vec{v}_0=\vec{a}\times m\vec{v}$

$v_0b=va$ （在最近处， $\vec{v}\perp\vec{a}$ ）

将 $v=\dfrac{b}{a}v_0$ 代入能量守恒式，得：

$a=\sqrt{b^2+\dfrac{c}{v_0^2}}$

4. 解：自由度为一，取θ为广义坐标

设棒长为$2l$，

OA棒重心$y_1 = l\sin(\theta+\alpha)$

$\because$ $\overline{OA} = \overline{AB}$

$\therefore$ $\angle ABO = \angle AOB$

$\therefore$ AB与ox轴成$\alpha-\theta$

故AB棒重心$y_2 = 2l\sin(\theta+\alpha)+l\sin(\alpha-\theta)$

虚功原理：$\delta w = 0$

$$\begin{aligned}\delta w &= mg\delta y_1 + mg\delta y_2 \\ &= mgl\cos(\theta+\alpha)\delta\theta + 2mgl\cos(\theta+\alpha)\delta\theta - mgl\cos(\alpha-\theta)\delta\theta \\ &= mgl[3\cos(\theta+\alpha)-\cos(\alpha-\theta)]\delta\theta = 0\end{aligned}$$

$\therefore$ $3\cos(\theta+\alpha)-\cos(\alpha-\theta)=0$

得：$tg\theta = \frac{1}{2}ctg\alpha$

5. 解：

$$\begin{cases} x_1 = l\sin\theta_1 \\ y_1 = l\cos\theta_1 \\ x_2 = l\sin\theta_1 + l\sin\theta_2 \\ y_2 = l\cos\theta_1 + l\cos\theta_2 \end{cases}$$

$$\begin{aligned}T &= \frac{1}{2}m(\dot{x}_1^2+\dot{y}_1^2)+\frac{1}{2}m(\dot{x}_2^2+\dot{y}_2^2) \\ &= \frac{1}{2}ml^2\dot{\theta}_1^2+\frac{1}{2}m[l^2\dot{\theta}_1^2+l^2\dot{\theta}_2^2+2l^2\cos(\theta_1-\theta_2)\dot{\theta}_1\dot{\theta}_2] \\ &\approx ml^2\dot{\theta}_1^2+\frac{1}{2}ml^2\dot{\theta}_2^2+ml^2\dot{\theta}_1\dot{\theta}_2\end{aligned}$$

$$\begin{aligned}V &= -mgl\cos\theta_1 - mg(l\cos\theta_1+l\cos\theta_2) \\ &= -2mgl\cos\theta_1 - mgl\cos\theta_2\end{aligned}$$

小振动，所以$\sin\theta\approx\theta$，$\cos\theta\approx 1-\frac{1}{2}\theta^2$

$\therefore$ $L \approx ml^2\dot{\theta}_1^2+\frac{1}{2}ml^2\dot{\theta}_2^2+ml^2\dot{\theta}_1\dot{\theta}_2-mgl\theta_1^2-\frac{1}{2}mgl\theta_2^2+3mgl$　$3mgl$为常数

代入拉格朗日方程，得：

$$\begin{cases} 2l\ddot{\theta}_1+l\ddot{\theta}_2+2g\theta_1=0 \\ l\ddot{\theta}_1+l\ddot{\theta}_2+g\theta_2=0 \end{cases}$$

设 $\theta_1 = A_1 e^{i\omega t}$, $\theta_2 = A_2 e^{i\omega t}$ 得:

$$\begin{cases} 2l\omega^2 A_1 + l\omega^2 A_2 - 2gA_1 = 0 \\ l\omega^2 A_1 + l\omega^2 A_2 - gA_2 = 0 \end{cases}$$

即 $\begin{pmatrix} 2l\omega^2 - 2g & l\omega^2 \\ l\omega^2 & l\omega^2 - g \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = 0$

存在非零解的条件:

$$\begin{vmatrix} 2l\omega^2 - 2g & l\omega^2 \\ l\omega^2 & l\omega^2 - g \end{vmatrix} = 0 \quad 即 \quad (2l\omega^2 - 2g)(l\omega^2 - g) = l^2\omega^4$$

得: $\omega = \sqrt{\dfrac{g(2 \pm \sqrt{2})}{l}}$

即: $\omega_1 = \sqrt{\dfrac{g(2+\sqrt{2})}{l}}$, $\omega_2 = \sqrt{\dfrac{g(2-\sqrt{2})}{l}}$

代回原方程 $\begin{cases} 2l\omega^2 A_1 + l\omega^2 A_2 - 2gA_1 = 0 \\ l\omega^2 A_1 + l\omega^2 A_2 - gA_2 = 0 \end{cases} \Rightarrow \begin{cases} 2(l\omega^2 - g)A_1 + l\omega^2 A_2 = 0 \\ l\omega^2 A_1 + (l\omega^2 - g)A_2 = 0 \end{cases}$

对于 $\omega_1 = \sqrt{\dfrac{g(2+\sqrt{2})}{l}}$, $l\omega_1^2 = (2+\sqrt{2})g$

$$\begin{cases} 2(2+\sqrt{2})gA_1 - 2gA_1 + (2+\sqrt{2})gA_2 = 0 \\ (2+\sqrt{2})gA_1 + (2+\sqrt{2})gA_2 - gA_2 = 0 \end{cases}$$

$$\begin{cases} 2(1+\sqrt{2})A_1 + (2+\sqrt{2})A_2 = 0 \\ (2+\sqrt{2})A_1 + (1+\sqrt{2})A_2 = 0 \end{cases}$$

$\Rightarrow \quad \dfrac{A_1}{A_2} = -\dfrac{\sqrt{2}}{2} = -\dfrac{1}{\sqrt{2}}$

对于 $\omega_2 = \sqrt{\dfrac{g(2-\sqrt{2})}{l}}$, $l\omega_2^2 = (2-\sqrt{2})g$

$$\begin{cases} 2(1-\sqrt{2})A_1 + (2-\sqrt{2})A_2 = 0 \\ (2-\sqrt{2})A_1 + (1-\sqrt{2})A_2 = 0 \end{cases}$$

$\Rightarrow \quad \dfrac{A_1}{A_2} = \dfrac{\sqrt{2}}{2} = \dfrac{1}{\sqrt{2}}$

对于 ω_1，$\begin{cases}\theta_1 = A_1^{(1)}\cos(\omega_1 t+\varphi_1)\\ \theta_2 = A_2^{(1)}\cos(\omega_1 t+\varphi_1)\end{cases} \Rightarrow \begin{cases}\theta_1 = A_1^{(1)}\cos(\omega_1 t+\varphi_1)\\ \theta_2 = -\sqrt{2}A_1^{(1)}\cos(\omega_1 t+\varphi_1)\end{cases}$

对于 ω_2，$\begin{cases}\theta_1 = A_1^{(2)}\cos(\omega_2 t+\varphi_2)\\ \theta_2 = A_2^{(2)}\cos(\omega_2 t+\varphi_2)\end{cases} \Rightarrow \begin{cases}\theta_1 = A_1^{(2)}\cos(\omega_2 t+\varphi_2)\\ \theta_2 = \sqrt{2}A_1^{(2)}\cos(\omega_2 t+\varphi_2)\end{cases}$

一般解

$$\begin{cases}\theta_1 = C_1A_1^{(1)}\cos(\omega_1 t+\varphi_1)+C_2A_1^{(2)}\cos(\omega_2 t+\varphi_2)\\ \theta_2 = -C_1\sqrt{2}A_1^{(1)}\cos(\omega_1 t+\varphi_1)+C_2\sqrt{2}A_1^{(2)}\cos(\omega_2 t+\varphi_2)\end{cases}$$

即 $\begin{cases}\theta_1 = A\cos(\omega_1 t+\varphi_1)+B\cos(\omega_2 t+\varphi_2)\\ \theta_2 = -\sqrt{2}A\cos(\omega_1 t+\varphi_1)+\sqrt{2}B\cos(\omega_2 t+\varphi_2)\end{cases}$ 其中 $A = C_1A_1^{(1)}$，$B = C_2A_1^{(2)}$

初始条件代入得:

$$\begin{cases}\theta_0 = A\cos\varphi_1 + B\cos\varphi_2\\ \theta_0 = -\sqrt{2}A\cos\varphi_1 + \sqrt{2}B\cos\varphi_2\\ -A\omega_1\sin\varphi_1 - B\omega_2\sin\varphi_2 = 0\\ \sqrt{2}A\omega_1\sin\varphi_1 - \sqrt{2}B\omega_2\sin\varphi_2 = 0\end{cases}$$

$$\Rightarrow \quad \begin{cases}\sin\varphi_1 = 0\\ \sin\varphi_2 = 0\end{cases} \Rightarrow \begin{cases}\varphi_1 = 0\\ \varphi_2 = 0\end{cases}$$

$$\therefore \quad \begin{cases}\theta_0 = A+B\\ \theta_0 = -\sqrt{2}A+\sqrt{2}B\end{cases} \Rightarrow \begin{cases}A = \dfrac{2-\sqrt{2}}{4}\theta_0\\ B = \dfrac{2+\sqrt{2}}{4}\theta_0\end{cases}$$

$$\therefore \quad \begin{cases}\theta_1 = \dfrac{2-\sqrt{2}}{4}\theta_0\cos\omega_1 t + \dfrac{2+\sqrt{2}}{4}\theta_0\cos\omega_2 t\\ \theta_2 = -\dfrac{\sqrt{2}-1}{2}\theta_0\cos\omega_1 t + \dfrac{\sqrt{2}+1}{2}\theta_0\cos\omega_2 t\end{cases}$$

6. 解: (1) 按流线微分方程 $\dfrac{dx}{-\dfrac{cy}{x^2+y^2}} = \dfrac{dy}{\dfrac{cx}{x^2+y^2}} = \dfrac{dz}{0}$ 得:

$$\frac{cxdx}{x^2+y^2} = -\frac{cydy}{x^2+y^2};\ \ dz = 0$$

$$\therefore \quad \frac{cd(x^2+y^2)}{x^2+y^2}=0; \quad dz=0 \quad \Rightarrow \quad z=\text{常数}$$

当 $x^2+y^2\neq 0$ 时，$x^2+y^2=$常数

所以流线方程为 $\begin{cases} x^2+y^2=C_1 \\ z=C_2 \end{cases}$ C_1, C_2 为常数

流线分布为同心圆

在 $x\neq 0,\ y\neq 0$ 处，

$$\omega_x=\frac{1}{2}(\nabla\times\vec{v})_x=\frac{1}{2}(\frac{\partial v_z}{\partial y}-\frac{\partial v_y}{\partial z})=0$$

$$\omega_y=\frac{1}{2}(\nabla\times\vec{v})_y=\frac{1}{2}(\frac{\partial v_x}{\partial z}-\frac{\partial v_z}{\partial x})=0$$

$$\omega_z=\frac{1}{2}(\nabla\times\vec{v})_z=\frac{1}{2}(\frac{\partial v_y}{\partial x}-\frac{\partial v_x}{\partial y})=\frac{1}{2}(c\frac{y^2-x^2}{(x^2+y^2)^2}-c\frac{y^2-x^2}{(x^2+y^2)^2})=0$$

旋度

$$\begin{aligned}\nabla\times\vec{v}&=(\vec{i}\frac{\partial}{\partial x}+\vec{j}\frac{\partial}{\partial y}+\vec{k}\frac{\partial}{\partial z})\times(v_x\vec{i}+v_y\vec{j}+v_z\vec{k})\\&=(\frac{\partial v_z}{\partial y}-\frac{\partial v_y}{\partial z})\vec{i}+(\frac{\partial v_x}{\partial z}-\frac{\partial v_z}{\partial x})\vec{j}+(\frac{\partial v_y}{\partial x}-\frac{\partial v_x}{\partial y})\vec{k}\\&=-\frac{\partial v_y}{\partial z}\vec{i}+\frac{\partial v_x}{\partial z}\vec{j}+(\frac{\partial v_y}{\partial x}-\frac{\partial v_x}{\partial y})\vec{k}\\&=0\end{aligned}$$

所以无旋

或者，转动角速度 $\vec{\omega}=\frac{1}{2}(\nabla\times\vec{v})=0$，无旋

散度

$$\begin{aligned}\nabla\bullet\vec{v}&=(\vec{i}\frac{\partial}{\partial x}+\vec{j}\frac{\partial}{\partial y})\bullet(v_x\vec{i}+v_y\vec{j})\\&=\frac{\partial v_x}{\partial x}+\frac{\partial v_y}{\partial y}\\&=\frac{\partial}{\partial x}(-\frac{cy}{x^2+y^2})-\frac{\partial}{\partial y}(\frac{cx}{x^2+y^2})\\&=\frac{-c(x^2+y^2)+2cxy+c(x^2+y^2)-2cxy}{(x^2+y^2)^2}\\&=0\end{aligned}$$

故流体不能压缩。

(2) 理想流体：①无粘滞性 ②不可压缩，ρ =常数

稳恒流体：$\frac{\partial \vec{v}}{\partial t}=0$

重力场：$\vec{f}=-\nabla V$，V 为体势能密度。

$\therefore$欧勒方程化为 $-\rho\vec{v}\times(\nabla\times\vec{v})+\nabla(\frac{1}{2}\rho v^2+P+V)=0$

沿某一流线积分，得

$$-\rho\int_{流线}\vec{v}\times(\nabla\times\vec{v})\cdot d\vec{r}+\int_{流线}\nabla(\frac{1}{2}\rho v^2+P+V)=0$$

$\because\ \vec{v}\times(\nabla\times\vec{v})\perp\vec{v}$,而 $\vec{v}\,\|\,d\vec{r}$

$\therefore\vec{v}\times(\nabla\times\vec{v})\cdot d\vec{r}=0$

有 $\int_{流线}\nabla(\frac{1}{2}\rho v^2+P+V)\cdot d\vec{r}=0$

$$\int_{流线}[\frac{\partial}{\partial x}(\frac{1}{2}\rho v^2+P+V)dx+\frac{\partial}{\partial y}(\frac{1}{2}\rho v^2+P+V)dy+\frac{\partial}{\partial z}(\frac{1}{2}\rho v^2+P+V)dz]=0$$

即 $\int_{流线}d(\frac{1}{2}\rho v^2+P+V)=0$

$\therefore\frac{1}{2}\rho v^2+P+V=$常数（该常数在固定流线上为常数，但对于不同流线，常数值可以不一样）

设$V=\rho U$，U 为单位质量势能，即质量势能密度。

可化成：$\frac{1}{2}\rho v^2+P+\rho U=$常数（沿流线）

这就是伯努力方程。

实际上：U=g h

$\therefore\frac{1}{2}\rho v^2+P+\rho gh=$常数　　(学生可以不写这部分)